

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION
Detailed Marking Report — Chemistry Exam-21

Seemab · Class 9 (Science Group) · Chapters 1 to 9 · Attempted 9 September 2026 · deep-check verified

41 / 46

TOTAL

89.1%

PERCENTAGE

A

GRADE

25 / 30

SECTION A

8 / 8

SECTION B

8 / 8

SECTION C

SECTION A — every question (25 / 30)

#	Question	Hers	Key	Mark	Note and source in her textbook
1	Which branch of chemistry mainly studies the energy changes that accompany chemical reactions?	A	A	1	Ch1 p.1 — explain that chemistry has many sub-fields. Source: Ch1, branches-of-chemistry chapter opener.
2	Which statement best distinguishes technology from science?	B	B	1	Ch1 p.1 — differentiate between 'science', 'technology' and 'engineering'.
3	A mixture whose particles are too small to settle but large enough to scatter light (Tyndall ef	C	C	1	Ch2 p.1 — identify solutions, colloids and suspensions as mixtures.
4	Which is an example of a solid solute dissolved in a liquid solvent forming a true solution?	D	D	1	Ch2 p.1 — identify solutions, colloids and suspensions and give an example of each.
5	Diamond, graphite and fullerenes are different forms of carbon. These are called:	B	B	1	Ch2 p.1 — explain the allotropic forms of solids (examples include diamond, graphite and fullerenes).
6	An atom as a whole is electrically neutral because:	A	A	1	Ch3 p.1 — state the relative charge of subatomic particles; interpret the relationship between a subatomic particle, its mass and charge. Source: Ch3 p.33, heading "Charg
7	An atom of zinc (atomic number 30, mass number 65). Number of neutrons?	D	D	1	Ch3 p.2 — determine the number of protons and neutrons of different isotopes. Source: Ch3 p.33, "No. of neutrons = mass number – atomic number".
8	Which statement correctly describes isotopes of the same element?	C	C	1	Ch3 p.2 — define isotopes as different atoms of the same element that have the same number of protons but different neutrons.
9	The Quantum Mechanical Model describes the electron's position in terms of:	B	A	0	Ch3 p.1 — state the electrons are quantum particles with probabilistic paths whose exact location cannot be mapped. Source: Ch3 p.31, "Quantum Mechanical Model".
10	Transition elements show variable oxidation states mainly because:	B	B	1	Ch4 p.2 — describe the transition elements as metals that have variable oxidation numbers. Source: Ch4 p.64, §4.5 "Transition Elements", point 3 "Variable Oxidation State
11	The LENGTH of a period is determined mainly by:	D	D	1	Ch4 p.1 — identify the group or period or block of an element using its electronic configuration. Source: Ch4 p.68, Key Points — "The length of a period in the periodic t
12	Group 1 and Group 2 elements are together called the:	C	C	1	Ch4 p.1 — identify the block of an element using its electronic configuration. Source: Ch4 p.68, Key Points — "Group 1 and 2 elements are called s-block elements, since s
13	Electron affinity is best defined as:	A	B	0	Ch4 p.1 — identify trends in groups and periods including electron affinity. Source: Ch4 p.68, Key Points — "Electron affinity is the amount of energy released when an el
14	Which transition metal is used as the catalyst in the Haber process?	B	A	0	Ch4 p.2 — describe transition elements as metals that act as catalysts for industrial purposes (example: Haber process). Source: Ch4 p.64, §4.5, point 5(a) — "Iron is use
15	Transition elements have high densities mainly because of their:	C	C	1	Ch4 p.2 — describe transition elements as metals that have high densities. Source: Ch4 p.64, §4.5, point 1 "High Density".
16	A covalent bond in which BOTH shared electrons come from only ONE atom is a:	D	D	1	Ch5 p.1 — define ionic, covalent, coordinate covalent and metallic bonds.
17	When potassium reacts with chlorine to form KCl, the potassium atom:	A	A	1	Ch5 p.1 — compare between the formation of cations and anions.
18	Metals conduct electricity well mainly because they contain:	B	B	1	Ch5 p.2 — justify the availability of free charged particles (electrons or ions) for conduction of electricity in metallic bonds.
19	Diamond has an extremely high melting point mainly because it:	A	C	0	Ch5 p.2 — interpret the strength of forces of attraction and their impact on melting/boiling points; justify the suitability of usage of diamond (e.g. in cutting tools).
20	One mole of any substance contains Avogadro's number of particles, which equals:	D	D	1	Ch6 p.1 — define mole as the amount of substance containing Avogadro's number (6.022×10^{23}) of particles.
21	Moles present in 2.3 g of sodium metal (Na = 23):	B	B	1	Ch6 p.1 — use the relationship amount of substance = mass / molar mass to calculate number of moles.
22	Which has an empirical formula DIFFERENT from its molecular formula?	A	A	1	Ch6 p.1 — define molecular formula and empirical formula of a compound.
23	Coefficients for $__ \text{H}_2\text{O}_2 \rightarrow __ \text{H}_2\text{O} + __ \text{O}_2$:	D	D	1	Ch6 p.1 — construct chemical equations to show reactants forming products, including balancing.
24	Oxidation number of sulfur in H_2SO_4 :	C	C	1	Ch7 p.1 — use roman numerals / rules to indicate the oxidation number of an element in a compound.
25	In a redox reaction, the oxidizing agent is the substance that:	B	B	1	Ch7 p.1 — identify oxidizing and reducing agents in redox reactions.
26	Oxidation number of chromium in the chromate ion, CrO_4^{2-} :	A	A	1	Ch7 p.1 — explain that the sum of the oxidation numbers in an ion is equal to the charge on the ion.
27	A catalyst speeds up a chemical reaction mainly by:	B	C	0	Ch8 p.1 — explain that activation energy depends on reaction pathway, which can be changed using a catalyst.
28	In an exothermic reaction, the enthalpy change (ΔH) is:	D	D	1	Ch8 p.1 — state that thermal energy is called enthalpy change and recognize its sign as negative for exothermic and positive for endothermic reactions.
29	A reaction that proceeds both forward and reverse under the same conditions and never completes	A	A	1	Ch9 p.1 — recognize that reversible reactions are shown by $\rightleftharpoons$ and may not go to completion.
30	Colour change when hydrated $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ is heated	B	B	1	Ch9 p.1 — describe the effect of heat on hydrated compounds, particularly

	and loses water of crystallization:			cobalt(II) chloride. Source: Ch9 Key Points, p.138 — "Anhydrous cobalt(II) chloride is a blue so
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Red rows lost the mark; amber rows are judgement calls. Every letter was re-read from the scan and cross-checked by an independent deep-check pass.

41 out of 46 · 89.1%	16/16 Sections B + C	83% MCQs
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The short version. Seemab scored 41 out of 46, her strongest paper since August. Her written work was **perfect** — both short questions and all six parts of the long numerical, sixteen marks out of sixteen. All five lost marks are multiple choice, and all five are specific facts rather than method failures.

Two results matter more than the score. First, the habit that has cost her marks on six consecutive papers — doing the mathematics and then not writing the concluding sentence — **cost her nothing here**. Every part of every question ends with an explicit, labelled final answer. Second, her multiple choice climbed from 60% on Monday's maths paper to 83% here, the first movement since that weakness was identified.

The five MCQs she missed

Q	Topic	Her answer	Correct	The error
9	Quantum mechanical model (Ch3)	Fixed circular orbits	Probability regions called orbitals	She gave the Bohr model. The quantum mechanical model replaced fixed orbits with regions of probability.
13	Electron affinity (Ch4)	Energy needed to remove the outermost electron	Energy released when a gaseous atom gains an electron	That is the definition of ionization energy . The two are opposites: one removes an electron, the other gains one.
14	Haber process catalyst (Ch4)	Nickel	Iron	Straight recall. Nickel is the catalyst for hydrogenating alkenes, not for ammonia.
19	Why diamond's melting point is so high (Ch5)	Small individual molecules held by weak forces	Giant covalent (network) structure	She described a simple molecular solid, which would melt easily. Diamond is one continuous covalent network, which is exactly why it does not.
27	How a catalyst works (Ch8)	Increases the overall enthalpy change	Provides a pathway with lower activation energy	A genuine misconception worth correcting directly: a catalyst never changes ΔH . It lowers the barrier, not the energy difference between reactants and products.

Questions 13 and 14 look transposed against the key, which is the signature of a mis-numbered answer list. It was formally tested: her answers score 25 matches at the correct alignment, against 2, 1, 2 and 0 at shifts of -2, -1, +1 and +2. There is no misalignment; these are two independent errors that happen to look like a swap.

Section B — 8/8

Q	Topic	Mark	Comment
2(i)	Burning hydrogen	4/4	2H ₂ (g) + O ₂ (g) → 2H ₂ O(l), correct coefficients and state symbols, classified exothermic, with the reason given in terms of bond breaking absorbing energy and bond forming releasing more. Never set in any of the 20 earlier chemistry papers.
2(ii)	Aluminium sulfate	4/4	LCM of 3 and 2 = 6, giving 2 and 3, ratio 2:3, criss-cross to Al ₂ (SO ₄) ₃ with correct bracket notation. Also never set before.

Section C — 8/8

A six-part numerical chain on an unknown dicarboxylic acid, every step correct:

Part	Required	Her answer	Mark
(a)	Empirical formula from 49.3% C, 6.8% H, 43.9% O	C ₃ H ₅ O ₂	3/3
(b)	Molecular formula given M _r = 146	C ₆ H ₁₀ O ₄	1/1
(c)	Moles in a 43.8 g sample	0.300 mol	1/1
(d)	Number of molecules	1.807 × 10 ²³	1/1
(e)	Balanced combustion equation	2C ₆ H ₁₀ O ₄ + 13O ₂ → 12CO ₂ + 10H ₂ O	1/1
(f)	Mass of CO ₂ from 0.3 mol	79.2 g	1/1

What this paper shows

1. The missing-conclusion habit did not cost her anything. That streak ran six papers — Maths exam-10, exam-13, exam-15, exam-16 twice, exam-17 three times, Chemistry exam-19 and exam-20. Here every sub-part ends with an explicit, correctly labelled final

answer: the classification and reason are stated, the bracketed formula is written out, and each of the six numerical parts closes with a value and its unit. This was the single highest-value change available to her and she has made it. **Tell her this.**

2. Her multiple choice moved for the first time. 60% on maths exam-17 on Monday, 83% here. One paper in a different subject is not proof, but the advice to write rough working rather than answer mentally went out with this paper, and the number went up. Worth reinforcing while the link is fresh.

3. All five losses are recall, not reasoning. None of them is a method failure or a calculation slip. Two are definitions confused with their opposite (electron affinity vs ionization energy; simple molecular vs giant covalent), one is a model she has mixed up with an older one, one is a named catalyst, and one is a genuine misconception about what a catalyst does to ΔH . These are learnable in an evening with flashcards.

One thing to mention gently

In part (c) her written working shows the numerator as **43.6**, while the question gives 43.8 g. Her final answer of 0.300 mol is correct and only follows from 43.8, so she used the right number and mis-copied it onto the page. No mark was affected. It is worth a word because a mis-copied figure usually does not stay harmless — here it happened not to propagate.

Similarly, in a charge check she volunteered on the aluminium sulfate question she wrote $2(+3) + 2(-2)$ where the second coefficient should be 3, and still concluded $6 + (-6) = 0$. No marks were deducted, since the question asked only for the LCM step, the formula and the brackets, and all three were already correct and complete before this bonus line. But she is not yet reliably separating a formula coefficient from the ion's own charge when checking under time pressure.

Revision priorities

- **A catalyst does not change ΔH .** It provides a lower-activation-energy pathway. This is the one item on the paper that is a true misconception rather than a forgotten fact.
- **Electron affinity versus ionization energy.** Learn them as an opposing pair, not separately: one is energy released on gaining an electron, the other energy needed to remove one.
- **Bonding and structure determine melting point.** Giant covalent networks (diamond, silicon dioxide) melt very high; simple molecular solids melt low.
- **The quantum mechanical model versus Bohr's.** Orbits were replaced by orbitals, which are probability regions.
- **Iron is the Haber process catalyst.** Straight memorisation.

Marking was independently deep-checked. All 30 MCQ answers were re-derived from the printed question stems rather than the answer key, and no key error was found. Her answer list was re-read from the scan letter by letter, and a shift guard was run across five alignments. Both passes reached 41/46 with no disagreements.